

Vantage Fleet OS

Design, Build and Delivery of the Maintenance, Dispatch and Compliance Platform

Submitted to	Vantage Freight Holdings Ltd.	Attention	Renata Okafor
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Proposed duration	34 weeks, seven phases	Validity	90 days from submission

1. Executive summary

We propose to design, build, test and deploy the Vantage Fleet OS platform in full: all fifty-seven functional requirements, both client applications, the Partner API and the data importers, across thirty-four weeks in seven accepted phases.

Vantage requires one system to carry a load from intake through assignment, pre-trip inspection, dispatch, post-trip reconciliation, maintenance triage and compliance review, with the workflow held as configuration rather than compiled into the application. The requirement set is large but, with three exceptions, conventional. Those exceptions carry nearly all of the technical risk, and the RFP identifies them plainly in clause 5.2: the data-driven workflow engine, the hours-of-service reconciliation ledger, and the eligibility rules that must return identical answers on the server and inside the driver's mobile application.

Our plan retires those three risks in the first half of the schedule rather than spreading them across it. The workflow engine and audit trail are accepted by week ten; eligibility, outcome routing and the mobile foundation by week seventeen. The hours-of-service ledger, the requirement with the strongest correctness obligation, is then built against a frozen configuration schema rather than a moving one.

PROTOTYPE EVIDENCE

We have built a working prototype of the three high-risk mechanics against this specification: the configurable status and transition engine with flag-based column resolution, four-state eligibility determination, a re-derived hours-of-service ledger, inspection outcome routing with override capture, and the immutable audit trail carrying eligibility snapshots. It carries thirty-three service-level tests and a transport-level suite including cross-company isolation checks. The estimates and risk positions below are drawn from that build rather than from analogy, and it is available for demonstration during evaluation.

Two findings from that work shape this proposal. Resolving workbench columns by status flag rather than by name removes risk R-3 by construction instead of by documentation, and costs nothing when done from the outset. And the reliable answer to risk R-2, drift between the mobile and server eligibility implementations, is not shared test fixtures but a single compiled rules package both runtimes import, which is achievable only if server and mobile share a language. That is the principal reason for the stack proposed in section 3.

2. Scope of work

Every functional requirement group in clause 4 is delivered and accepted in the phase shown. We comply in full with FR-1 to FR-57 with three qualifications: FR-50 ships as the reporting placeholder the RFP itself specifies, FR-56 fuel analytics is optional and offered as a follow-on item, and against FR-27 we exceed the requirement by delivering one shared implementation rather than two kept in synchronisation.

TABLE 1. FUNCTIONAL SCOPE BY PHASE

REQUIREMENTS	CAPABILITY AREA AND DELIVERED CONTENT	PHASE
FR-1 to FR-7	Access control, authentication, users. Six roles with platform-admin flag; company scoping and capability checks at the query layer; impersonation with safeguards; invitation onboarding; user administration; device-bound mobile tokens.	P1
FR-8 to FR-16	Records and catalogue. Carrier companies, vehicles with status and source marker, driver records with computed eligibility; vehicle types and interval templates; fault and defect catalogues with per-type exclusions; aliases and legacy mappings.	P1
FR-17 to FR-21	Loads and documents. Load entity with the VFO reference; status-gated list with per-status columns, route grouping and persistent bulk selection; detail view with document gallery and dwell timeline; the created-status edit lock.	P1-P2
FR-23 to FR-25 FR-30 to FR-35	Workflow engine and audit. Configurable statuses and transitions with all flag, role-visibility and field-set configuration; reorder, archive, flowchart and role matrix; advance, rank-enforced revert, bulk and whole-route operations; the immutable audit trail with captured inspection and eligibility data.	P2
FR-22, FR-26 to FR-29	Assignment, eligibility, routing. Cascading assignment filtered by eligibility with a live determination panel; ruleset resolution in priority order; outcome computation and auto-routing with mandatory override reason; the shared Record and Advance modal.	P3
FR-36 to FR-39	Workbenches, routes, triage. Maintenance and Dispatch boards with faceted filters, live search and in-place updates; route creation with auto-advance and derived stage badges; per-vehicle fault triage with severity breakdown.	P3
FR-40 to FR-43	Hours of service and compliance. The per-driver-cycle reconciled ledger with per-entry on-duty override; compliance workbench with declined-vehicle routing back to maintenance; record lifecycle and generated summaries.	P4
FR-44 to FR-48 FR-52	Bulk upload and integrations. Standard importer with per-row validation and historical hours verification; legacy importer with add-alias-and-revalidate; telematics web view and ingestion; the bearer-token Partner API and its documentation.	P5
FR-49, FR-51 FR-53 to FR-55	Self-service and administration. Dashboard builder with nine widget and six chart types; help centre and articles; per-role navigation builder with its non-widening resolver; notification templates; settings store.	P6
FR-5, FR-27	Mobile driver application. Assignment confirmation, pre-trip and post-trip inspection and duty-status entry; offline-first capture with conflict-safe synchronisation; device-bound tokens and remote wipe.	P3-P6

We scope to the exclusions at clause 11.2: billing logic, with mileage and stop data exported only; business intelligence dashboards; internationalisation; payment processing; public interfaces beyond the telematics Partner API; and automated transmission of compliance summaries, which are generated for export only. The temporary tooling of FR-57 is built for staging and removed from the production image at build time rather than hidden behind a runtime flag.

3. Technology stack

Clause 5.1 permits an equivalent stack with justification and names a TypeScript alternative as acceptable. We propose TypeScript across server, web and mobile, on PostgreSQL.

TABLE 2. PROPOSED STACK

LAYER	TECHNOLOGY	BASIS FOR SELECTION
Language	TypeScript on Node 22 LTS	One language across all three runtimes, which is what makes the shared rules package possible
Server	Express with a thin services layer	Matches the services-layer architecture of clause 5.2
Database	PostgreSQL 16 with Prisma	Typed schema, reviewable migrations; partial unique indexes enforce workflow invariants
Web client	React with Vite	In-place board updates without full-page reloads
Mobile	React Native, iOS and Android	Mandatory under clause 5.2; shares the rules package with the server
Offline store	SQLite with an append-only sync queue	Conflict-safe capture under NFR-6
Jobs and cache	BullMQ on Redis	Import processing, notifications, session store
Object storage	AWS S3	Load and inspection documents per clause 5.4
Authentication	Argon2id, HttpOnly cookies, device-bound refresh tokens	Clause 5.5 password and session requirements
Charts and sheets	Chart.js; ExcelJS with macros disabled	FR-49 chart types; safe parsing for FR-44 to FR-46
Deployment	Docker multi-stage non-root on EC2 behind an Application Load Balancer, GitHub Actions	Rolling instance replacement across the auto-scaling group gives zero-downtime releases with migrations applied on release
Testing	Vitest, Supertest, Playwright, Detox	The four test layers required by clause 5.6

3.1 Justification for departing from the reference stack

We do not claim the functional set is easier in Node; the RFP is correct that the engine, the ledger and the dual eligibility rules carry effort regardless of stack. The justification rests on one point. FR-27 requires eligibility logic to behave identically server-side and in the mobile app, and R-2 rates drift between those implementations medium-likelihood and high-impact with a single source of truth as the preferred mitigation. With a Python server and a JavaScript mobile app that is unattainable, and the mitigation degrades to fixtures that detect drift after it occurs. With one language, the rules compile to a single package the server imports and the mobile app bundles, so no second implementation exists to drift. If Vantage intends to hold Python capability for long-term ownership, that is a legitimate reason to prefer the reference stack and we are equally able to deliver on Django; R-2 then reverts to a monitored risk with contract tests executed against both implementations.

3.2 Forward roadmap

The following sit outside core scope and are neither scheduled nor priced here. They are recorded so the delivered architecture does not preclude them: a reporting engine over a read replica replacing the FR-50 placeholder; regulatory portal submission adapters for the summaries excluded at clause 11.2; additional telematics providers as further adapters behind the normalisation seam delivered in P5; change-data-capture into a columnar warehouse for business intelligence, using the immutable audit rows as the event stream; predictive maintenance from retained telematics history; route optimisation and mapping over the existing route and stop entities; single sign-on alongside password authentication; and the FR-56 fuel analytics model.

4. System architecture

A thin transport and authorisation layer sits above a services layer holding the domain logic, with the workflow engine driven entirely by configuration rows. The shaded band in Figure 1 is the shared rules package described above.

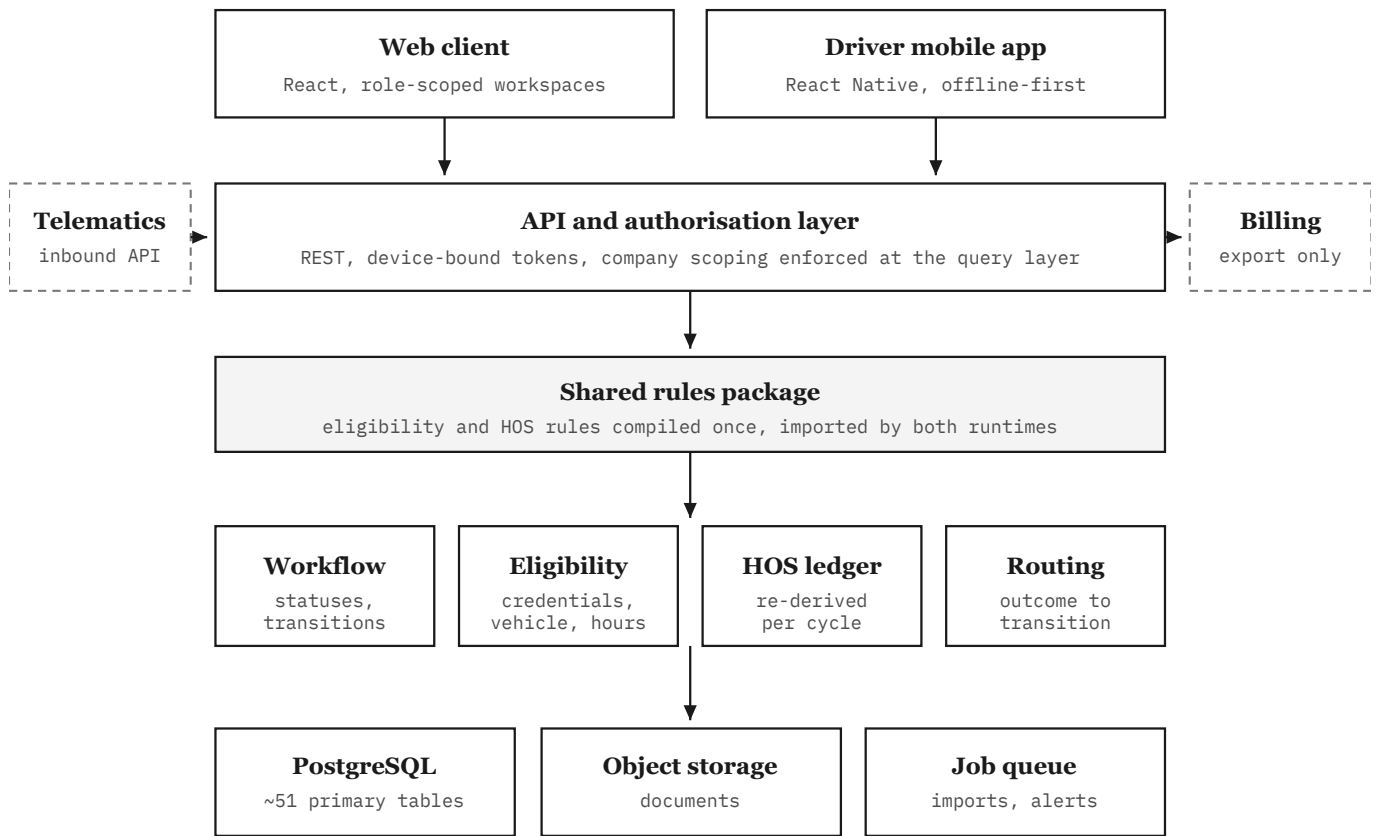


Figure 1. Layered architecture. The mobile application bundles the same rules package the server imports, so an eligibility determination made in the cab matches the one the server produces on synchronisation. The server remains authoritative and revalidates every submission.

The data model comprises approximately fifty-one primary tables across six domains, consistent with Appendix A: loads and workflow; the reference catalogue and legacy mappings; telematics events and duty-status changes; compliance records, routes and carrier companies; configuration and content; and framework tables. Two rules hold throughout. Derived values are never stored, so route stage, eligibility and hours availability are computed on read and cannot go stale. Captured audit data is held as one versioned JSON column on the status log rather than a set of nullable snapshot keys, keeping each audit row immutable and self-contained.

4.1 Dispatch workflow and outcome routing

Figure 2 shows the default configured workflow. Every box is a row in the status table and every arrow a row in the transition table, so the shape shown is configuration rather than code and an administrator can alter it without a release.

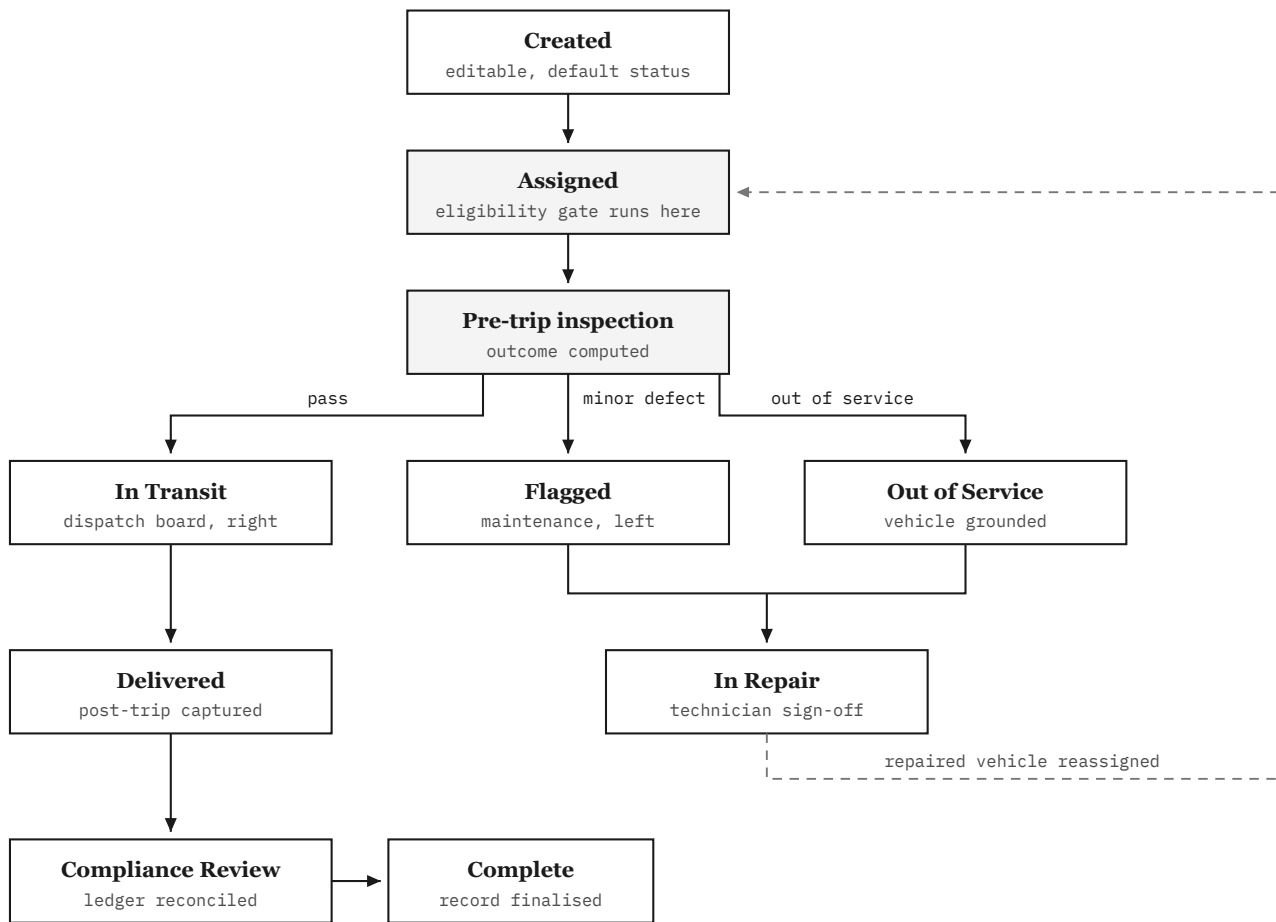


Figure 2. Default dispatch workflow. Solid arrows are configured transitions and the dashed arrow is the repaired-vehicle return path. The three labelled branches leaving the inspection stage are selected automatically by the computed inspection outcome.

FR-28 specifies the outcome priority exactly and we implement it as a pure function, testable without a database: any out-of-service defect makes the outcome out-of-service; failing that, any minor defect makes it a minor defect; otherwise the inspection passes. The computed outcome selects the transition whose trigger matches. A manual override bypasses that selection and requires a reason, validated before any write, so an override without a reason cannot leave a partial record. Submitting an inspection and advancing the load occur in one transaction, leaving no window in which an inspection exists without the status change it caused. Workbench columns resolve by status flag only, so renaming a status cannot break the Dispatch or Maintenance board; that rename is an automated regression test rather than a documented caveat, which is the improvement invited by the note under clause 4.11.

5. Approach

Each phase concludes with a deployed increment on an agreed environment, a passing integration run, and a demonstration against the requirements that phase claims, which is the acceptance definition at clause 6.3. Within a phase we work in one-week iterations with a mid-phase review, so a misread requirement surfaces in days rather than at the handover checkpoint. Requirements are tracked by FR number from proposal to acceptance, so the status of each of the fifty-seven items is visible to both parties throughout.

Scope control carries more weight here than usual because clause 11.3 rates creep on the configurable engine high for both likelihood and impact. Our control is a configuration schema freeze at the end of phase two, after which additions to the status, transition or field-set schema are change control rather than refinement. A configurable system invites indefinite configurability, and this freeze is what stops phase two consuming phase four.

The three high-risk mechanics

- **Workflow engine.** Statuses carry an explicit position and direction is enforced by rank rather than by which edges happen to exist: advance may move only to an equal or later position, revert only to an earlier one, making the backward-only rule of FR-31 structural. Two partial unique indexes make ambiguous auto-routing impossible at the database level, permitting one default target per source status and one transition per outcome per source status.
- **Hours-of-service ledger.** Availability is re-derived from duty-status entries on every read and never cached, as R-6 requires. The single-reconciled-ledger invariant is a property test over generated duty sequences covering reset boundaries and overlapping assignments, not example cases alone.
- **Dual eligibility.** One compiled rules package imported by both runtimes. The mobile application calls the same function the server calls; the server remains authoritative and revalidates on submission.

6. Milestones and timeline

Thirty-four weeks, within the twenty-four to forty week band indicated at clause 7. Phases run in sequence because each depends materially on its predecessor, with the mobile application as a parallel stream from week eleven once the workflow engine is stable enough to build against.

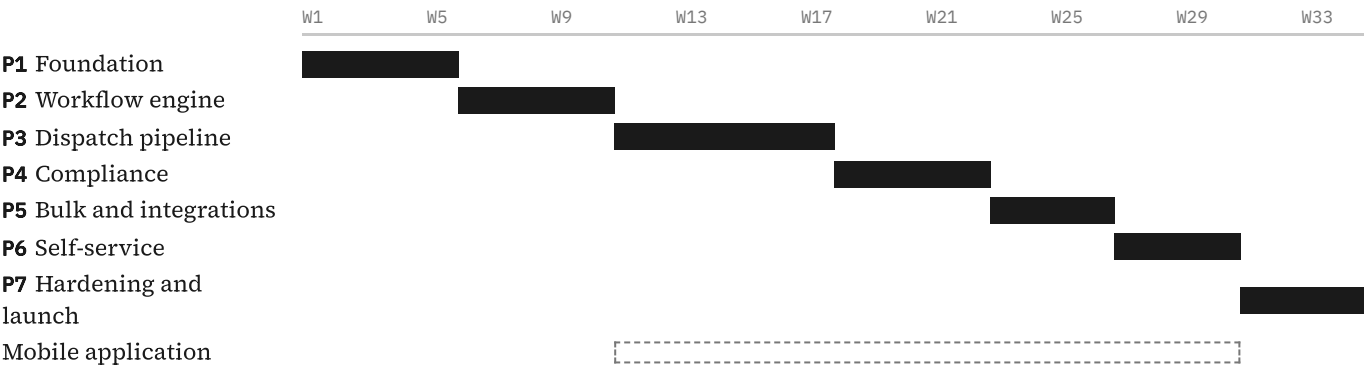


TABLE 3. PHASE DELIVERABLES AND ACCEPTANCE

PHASE	WEEKS	HANDOVER INCREMENT AND ACCEPTANCE DEMONSTRATION
P1	1–5	Foundation. Pipeline, container image and deployment; authentication, users, invitations, impersonation, access control; carriers, vehicles, drivers, catalogue; basic loads. <i>Accepted when a scoped user cannot reach another company's data through any route, demonstrated at the query layer.</i>
P2	6–10	Workflow engine. Configurable statuses and transitions; advance, revert and bulk variants; immutable audit log; workflow diagram and role matrix. <i>Accepted when a status and transition added through the interface change load behaviour with no deployment. The configuration schema freezes here.</i>
P3	11–17	Dispatch pipeline. Assignment with live eligibility; routes; Maintenance and Dispatch workbenches; shared Record and Advance modal; mobile foundation. <i>Accepted when identical eligibility inputs return identical verdicts from the server API and the mobile bundle.</i>
P4	18–22	Compliance. Hours-of-service ledger and allocation; compliance workbench and record lifecycle; generated summaries. <i>Accepted when cumulative on-duty totals reconcile against a driver's declared logs across a full cycle including qualifying resets.</i>
P5	23–26	Bulk and integrations. Standard and legacy importers; telematics verification and web view; Partner API and documentation. <i>Accepted when a five-hundred-row import validates in under ten seconds with leading zeros preserved and descriptive per-row errors.</i>
P6	27–30	Self-service and administration. Dashboard builder; navigation configuration; help centre; notification templates; settings. <i>Accepted when navigation configuration can reorder, rename and hide items for a role but cannot grant access that role does not hold.</i>
P7	31–34	Hardening and launch. Security and performance hardening; full end-to-end coverage including mobile offline sync; documentation; production deployment; UAT; knowledge transfer. <i>Accepted when NFR-1 to NFR-7 are met under representative load and a zero-downtime rolling deployment with migrations on release is demonstrated.</i>

As clause 6.4 invites, we can alternatively deliver a leaner first release comprising phases one to three with a single-ruleset compliance review, reaching a usable dispatch and maintenance system in approximately nineteen weeks, deferring dashboards, navigation configuration, the help centre, the legacy importer and the Partner API. This is a resequencing rather than a reduction in eventual scope.

7. Risk, quality and security

TABLE 4. RESPONSE TO THE RISK REGISTER AT CLAUSE 11.3

REF	RISK	OUR POSITION
R-1	Telematics API changes break ingestion	A normalisation seam ahead of ingestion absorbs payload changes in an adapter. Shapes are pinned and contract-tested against the Vantage sandbox.
R-2	Mobile and server eligibility drift	Eliminated, not mitigated: one compiled rules package imported by both runtimes. Shared fixtures still run against both call sites as a guard.
R-3	Renamed statuses break workbenches	Columns resolve by flag, never by name, with a rename regression test. No name dependency exists to document.
R-4	Scope creep on the configurable engine	Configuration schema freezes at the P2 handover; later schema additions are change control. The schedule depends on this holding.
R-5	Spreadsheet import edge cases	Per-row validation with descriptive errors and leading-zero normalisation runs before any write; malformed files fail at parse, not mid-transaction.
R-6	Hours-of-service reconciliation errors	Availability always re-derived server-side and never cached; the ledger invariant is property-tested over generated duty sequences.
R-7	Legacy data migration	A migration rehearsal in P5 runs the legacy importer against a full production extract before cutover, surfacing alias and mapping gaps against real data.
R-8	Mobile offline sync conflicts	Conflict rules settled in P3 before offline capture is built; entries are append-only with client-generated identifiers, making sync idempotent.

Non-functional requirements. Board and list queries run against covering indexes with eager loading so no page issues per-row queries, and the NFR-1 budgets are asserted in the pipeline against a seeded dataset so a regression fails the build. Stateless workers behind a load balancer with sessions in Redis meet NFR-2, load tested at eighty internal and two hundred and fifty mobile sessions in phase seven. Rolling zero-downtime deployments and daily backups with a rehearsed restore meet NFR-3. The browser matrix of NFR-4 is covered in the end-to-end suite. WCAG 2.1 AA contrast, keyboard-navigable primary flows and labelled controls are audited automatically and by manual traversal of the dispatch and inspection paths for NFR-5. Offline capture writes to a local append-only queue with client-generated identifiers, so replay on reconnection is idempotent and lossless for NFR-6. Every status change writes one immutable log row carrying actor, timestamps, statuses, captured inspection data, eligibility snapshot, stop count and flags, never updated or deleted by application code, for NFR-7.

Security. Authorisation is enforced in the services and query layer rather than the interface, with company scoping applied as a query predicate so a missing filter fails closed. Passwords use Argon2id at a tuned work factor with a six-hour reset window; session cookies are HttpOnly, Secure and SameSite; cross-site request forgery protection is enabled everywhere except the Partner API namespace; TLS is forced in production. Partner tokens are held per partner in encrypted configuration and compared in constant time. Secrets appear in neither source nor logs. Input is validated server-side, queries are parameterised, spreadsheets parse with macros disabled, and uploads are validated on content rather than extension. Mobile offline data sits in encrypted local storage behind device-bound tokens with remote wipe.

8. Commercial terms and assumptions

We offer a defect-fix warranty of sixty days from production go-live, exceeding the thirty-day minimum at clause 10.1. Ongoing support is offered against four severity levels, from one-hour response and same-day resolution for a production outage through to next-release handling of cosmetic defects, and we will align hours and targets with the values Vantage completes at clause 10.2. All source, deliverables, documentation and associated intellectual property are owned by Vantage and developed in a Vantage-controlled repository

from the first commit rather than transferred at completion; every third-party component is disclosed with its licence, and we incorporate no pre-existing vendor intellectual property that is not licensed to Vantage on a perpetual, royalty-free basis. Handover delivers architecture and data-model documentation, deployment and operational runbooks, Partner API documentation and an administrator configuration guide covering the workflow engine, the ledger and the settings store, followed by knowledge-transfer sessions run against the live system.

Consistent with clause 11.1 we assume Vantage provides telematics sandbox access, the SMS gateway and object storage account, reference and seed data, and a production hosting environment unless we are separately engaged for managed hosting; that a purchased administration theme is acceptable and bespoke visual design is a follow-on item; that rulesets are expressed per applicable federal exemption category; and that staff are available for phase reviews and acceptance testing within one week of each handover, failing which the affected phase boundary moves rather than the following phase compressing. The functional requirements in the RFP are treated as authoritative where they conflict with the reference implementation.

This response addresses every section of the RFP and follows the response checklist at clause 12. Company particulars, insurances, qualification case studies and client references required by VQ-1 to VQ-10 accompany this document under separate cover. It contains no pricing, rate cards, fixed fees or budget figures, per clauses 3.9 and 12, and remains valid for ninety days from submission.